AKIRA 1500 cc engine

2026 · design in progress

Engineering problem and work

Contributed to a team concept for a 1500 cc inline-four motorcycle engine, focusing on valvetrain trade-offs, DOHC configuration, electronically controlled variable valve timing, and ideal Otto-cycle analysis.

Process

Requirements → weighted decisions → thermodynamic model → design integration

Tools

DOHC · VVT · Otto cycle · Design trade-offs

Scope and status

80 hp at 5,000 rpm and 35% efficiency are design targets, not verified outcomes

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Portfolio case-study brief based on coursework and user-provided descriptions. This is NOT a copy of the original laboratory report. Diagrams on the website are explanatory illustrations, not photographs or original experimental plots.

Original source (may require Google Drive permission):
https://docs.google.com/document/d/1GOOLfW3Jkrx8u1mZNXE7TkX0c-HUWY4bEUoEcQ4meaI/edit